TT-D-301b
APRIL 5, 1960
SUPERSEDING
Int. Fed. Spec. TT-D-00301a (GSA-FSS)
August 10, 1959, and
Fed. Spec. TT-D-301
November 4, 1947

FEDERAL SPECIFICATION

DIBUTYL PHTHALATE: PLASTICIZER (FOR USE IN ORGANIC COATINGS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1. Scope. — This specification covers a dibutyl phthalate for use in protective coatings.

1.2 Classification.

1.2.1 *Grade and type.*—The dibutyl phthalate covered by this specification shall be of one type and one grade satisfactory for use as a plasticizer in organic coatings.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

PPP-B-636—Box, Fiberboard.
PPP-C-96 — Cans, Metal, 28 Gage and Lighter.
PPP-D-729—Drums: Metal, 55 Gallon (For Shipment of Noncorrosive Material).
PPP-D-760—Drums and Pails, Metal (5 and 16.64 Gallon).

Federal Standards:

Fed. Std. No. 102 — Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:

MIL-STD-129 — Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.—The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

ASTM Standards — Part 4.

(Copies may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Quantitative requirements. — The dibutyl phthalate shall conform to the requirements of table I.

TABLE I.—*Quantitative requirements*

Characteristics	Requirements	
	Minimum	Maximum
Specific gravity at 20°/20°C.	1.046	1.050
Ash (percent by weight)	---	0.01
Acidity (mg. of KOH per gram of sample)	---	0.067
Refractive index at 20°C.	1.490	1.492
Ester content (percent by weight)	99	---
Flash point	171°C.	---
Color (platinum cobalt scale)	---	No. 20

3.2 Qualitative requirements.

3.2.1 *Appearance.*—The dibutyl phthalate shall be clear and free from sediment and suspended matter when examined by transmitted light.

3.2.2 *Odor.*—The dibutyl phthalate shall have a mild characteristic odor.

3.2.3 *Water.*—The dibutyl phthalate shall be miscible without turbidity with 19 volumes of 10° heptane at 20°C.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling and inspection.—Unless otherwise specified, sampling and inspection shall be in accordance with methods 1011, 1021, and 1031 of Federal Test Method Standard No. 141.

4.3 Test procedures.

4.3.1 Tests shall be conducted in accordance with the applicable methods of Federal Test Method Standard No. 141, and as specified hereinafter.

Test	Method
Appearance	4261
Specific gravity	4183
Ash	5262
Acidity	5252
Refractive index	4371
Ester content	5172
Flash point	4294
Odor	4401
Water	4083

4.3.2 *Color.*—Color shall be determined by method ASTM 1209.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Level A.*—Unless otherwise specified, dibutyl phthalate shall be packaged in 1-quart

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or 1-gallon cans conforming to type V, class 4, oblong, of Federal Specification PPP–C–96; in 5-gallon drums conforming to type I, class 1, of Federal Specification PPP–D–760; in 55-gallon drums conforming to type II of Federal Specification PPP–D–729; or shipped bulk.

5.1.2 *Level C.*—Unless otherwise specified, dibutyl phthalate shall be packaged in containers conforming to the Consolidated Freight Classification Rules or other regulations applicable to such supplies.

5.2 Packing.

5.2.1 *Level A.*—Unless otherwise specified, twelve 1-quart or six 1-gallon cans shall be packed in a fiberboard box conforming to type I, W5c, style RSC of Federal Specification PPP–B–636. The 5 and 55-gallon drums require no further packing.

5.2.2 *Level B.*—Dibutyl phthalate shall be packed in accordance with 5.2.1 except that fiberboard boxes shall comply with type I, class 1, style RSC of Federal Specification PPP–B–636.

5.2.3 *Level C.*—Unless otherwise specified, dibutyl phthalate shall be packed in containers conforming to the Consolidated Freight Classification Rules or other regulations, for safe delivery to destination at the lowest applicable rate.

5.3 Marking.

5.3.1 *Civil agencies.* — Unless otherwise specified, shipments shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use of the material.—The dibutyl phthalate covered by this specification is intended for use as a plasticizer in organic protective coating materials. It has good solvent power for nitrocellulose, is compatible with the usual lacquer solvents and diluents, and imparts toughness and elasticity to the final coatings containing it.

6.2 Basis of purchase.—Dibutyl phthalate may be purchased either by volume, the unit being a U. S. gallon at 20°C. (68°F.) or by weight, the unit being one pound. The volume of deliveries may be determined by dividing the net weight, in pounds, by weight per gallon. To obtain the weight per gallon, multiply the specific gravity at 20°/20°C. (68°/68°F.) by 8.33. One gallon of dibutyl phthalate at this temperature weighs between 8.713 and 8.747 pounds.

6.3 Ordering data.—Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Level of packaging and packing required (see 5.1 and 5.2).

(*c*) Size of package and quantity required.

(*d*) Details of inspection and test records required (see section 4).

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—C E M O Q Sig
Navy—W
Air Force—

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